Course Code: IT602
Information Technology infrastructure
FinalTerm MCQ's file # 2 (Week 7-16)
Prepared by DL

1. **What is Network Function Virtualization (NFV)?**
   A. Virtualizing networking devices like firewalls and load balancers
   B. A method for data storage
   C. A type of network protocol
   D. A programming language

2. **What is the core layer in a layered network topology?**
   A. The center of the network
   B. The access layer
   C. The distribution layer
   D. The edge of the network

3. **What does the Spanning Tree Protocol (STP) do?**
   A. Prevents loops in the network
   B. Increases network speed
   C. Provides encryption
   D. Manages user access

4. **What is multihoming?**
   A. Connecting to multiple ISPs
   B. Using multiple routers
   C. Connecting multiple devices to a single ISP
   D. Using multiple network protocols

5. **What is Quality of Service (QoS)?**
   A. Ability to provide different data flow priority
   B. A type of network protocol
   C. A method for data storage
   D. A programming language

6. **What is a Virtual Tape Library (VTL)?**
   A. A storage device that uses disks for backups
   B. A physical tape storage system
   C. A type of RAID configuration
   D. A method for data encryption

7. **What does thin provisioning allow?**
   A. Allocating more storage capacity than is physically installed

B. Reducing storage costs
C. Increasing data redundancy
D. Enhancing data security

8. **What is the primary function of a Storage Area Network (SAN)?**
   A. To connect multiple servers to a central storage pool
   B. To provide internet access
   C. To manage network traffic
   D. To enhance user experience

9. **What is the purpose of a controller in storage systems?**
   A. To connect disks to a server
   B. To manage user access
   C. To encrypt data
   D. To compress data

10. **What is the main disadvantage of using tape storage?**
    A. High cost
    B. Fragility and slow access speed
    C. Limited capacity
    D. Complexity in management

11. **What is the main advantage of using blade servers?**
    A. Reduced wiring and lower deployment costs
    B. Higher processing power
    C. Increased physical size
    D. More moving parts

12. **What is the function of device drivers?**
    A. To connect hardware devices to the operating system
    B. To manage user access
    C. To perform calculations
    D. To store data

13. **What is the purpose of a shell in an operating system?**
    A. To provide a user interface for launching programs
    B. To manage hardware resources
    C. To store data
    D. To perform calculations

14. **What is the primary function of the kernel in an operating system?**
    A. To manage system resources and execute programs
    B. To provide user interfaces
    C. To connect hardware devices
    D. To store data

15. What is the main purpose of process scheduling?
A. To manage the execution of multiple processes
B. To allocate memory
C. To connect devices
D. To provide user interfaces

16. What is the main advantage of multi-core CPUs?
A. They can process multiple instructions simultaneously
B. They are cheaper than single-core CPUs
C. They consume more power
D. They are easier to program

17. What is virtualization availability?
A. The ability to restart virtual machines on different physical machines
B. The ability to run multiple applications on a single machine
C. The ability to manage user access
D. The ability to store data securely

18. What is the primary purpose of a systems management console?
A. To manage hypervisors and virtual machines
B. To provide user interfaces
C. To store data
D. To perform calculations

19. What is the main disadvantage of virtualization?
A. Introduction of an extra layer in the infrastructure
B. Increased hardware costs
C. Limited application support
D. Reduced processing speed

20. What is the purpose of a CPU cache?
A. To temporarily store data for faster access
B. To permanently store data
C. To manage user access
D. To connect hardware devices

21. What is the purpose of system calls in an operating system?
A. To provide an interface for applications to request services
B. To manage hardware resources
C. To store data
D. To perform calculations

22. What does RAID stand for?
A. Random Access Input Device
B. Redundant Array of Independent Disks
C. Reliable Architecture for Input Data

D. Remote Access Integration Device

23. What is the primary function of RAID?
A. To improve data redundancy and performance
B. To increase processing speed
C. To connect devices
D. To encrypt data

24. Which RAID level provides no redundancy?
A. RAID 1
B. RAID 5
C. RAID 0
D. RAID 10

25. What does RAID 1 do?
A. Mirrors data across multiple disks
B. Stripes data without redundancy
C. Combines striping and parity
D. Uses multiple parity blocks

26. Which RAID level uses parity for redundancy?
A. RAID 0
B. RAID 1
C. RAID 5
D. RAID 10

27. What does the term "hot-swappable" mean in storage systems?
A. Replacing a component without shutting down the system
B. A storage device with high temperature tolerance
C. A system that uses heat for operation
D. Replacing data in a RAID array

28. What is deduplication?
A. Eliminating duplicate copies of data
B. Encrypting data
C. Compressing data
D. Backing up data

29. What does a hypervisor do in virtualization?
A. Manages virtual machines
B. Increases hardware performance
C. Connects physical devices
D. Encrypts data

30. What is the difference between Type 1 and Type 2 hypervisors?
A. Type 1 is software-based; Type 2 runs directly on hardware

B. Type 1 runs directly on hardware; Type 2 runs on an operating system

31. What is live migration in virtualization?
A. Moving a virtual machine from one host to another without downtime
B. Creating a backup of a virtual machine
C. Increasing virtual machine speed
D. Encrypting virtual machines

32. What does the term "snapshot" mean in virtualization?
A. Capturing the state of a virtual machine at a specific moment
B. Copying data to another device
C. Increasing virtual machine performance
D. Encrypting virtual machine data

33. What does the acronym SDN stand for?
A. Software-Defined Networking
B. Security Device Network
C. System Data Node
D. Secure Data Networking

34. What is the main advantage of Software-Defined Networking (SDN)?
A. Centralized network control and programmability
B. Increased hardware cost
C. Reduced network security
D. Improved physical hardware design

35. What does Network Address Translation (NAT) do?
A. Converts private IP addresses to public IP addresses
B. Encrypts network traffic
C. Increases network bandwidth
D. Reduces latency in the network

36. What is the purpose of a firewall?
A. To monitor and control incoming and outgoing network traffic
B. To store data securely
C. To increase internet speed
D. To allocate IP addresses

37. What is the primary function of an intrusion detection system (IDS)?
A. To detect and alert on unauthorized access attempts
B. To encrypt sensitive data
C. To increase network bandwidth
D. To assign IP addresses

38. What is the difference between IDS and IPS?
A. IDS actively blocks threats; IPS only detects them

B. IDS detects threats, while IPS actively prevents them
C. IDS operates on hardware; IPS operates on software
D. IDS is faster than IPS

39. What is a honeypot in cybersecurity?
A. A decoy system designed to lure attackers
B. A high-security firewall
C. A type of encryption algorithm
D. A network protocol

40. What does VPN stand for?
A. Virtual Protection Network
B. Variable Private Network
C. Virtual Private Network
D. Verified Protocol Network

41. What is the primary purpose of a VPN?
A. To create a secure connection over the internet
B. To increase internet speed
C. To allocate IP addresses
D. To reduce hardware costs

42. What is a DDoS attack?
A. A distributed attack that overwhelms a server with traffic
B. An attack that decrypts sensitive data
C. A targeted malware attack
D. A method of data backup

43. What is the purpose of a penetration test?
A. To identify vulnerabilities in a system
B. To increase system performance
C. To encrypt sensitive data
D. To back up data

44. What is social engineering in cybersecurity?
A. A method for encrypting data
B. A way to configure firewalls
C. Manipulating people to disclose sensitive information
D. A tool for monitoring network traffic

45. What is the main goal of encryption?
A. To compress data
B. To back up data
C. To protect data by converting it into an unreadable format
D. To speed up data transmission

46. Fairness and Bias Mitigation in IT infrastructure primarily addresses:

A) Cost efficiency
B) Data collection and preparation
C) User interface design
D) Marketing strategies

47. Intellectual Property rights are important because they:

A) Encourage competition
B) Protect creations and innovations
C) Limit access to information
D) Increase software costs

48. What is a key consideration in the ethical deployment of AI and Automation?

A) Cost reduction
B) Human-Centered Approach
C) Market dominance
D) Data collection

49. Human-Centered Design (HCD) prioritizes:

A) Cost efficiency
B) User needs and experiences
C) Technical specifications
D) Market trends

50. Compliance and Legal Adherence in IT infrastructure management is crucial for:

A) Increasing profits
B) Data protection and privacy
C) Enhancing user experience
D) Reducing operational costs

Prepared by DL...
May You All Stay Success in your Whole Life,
Ameen! Summ Ameen...
Just Remember us in your Deep Prayers. JazakAllah…